Video Game Metadata, Rating, and Global Sales

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Abstract

This project analyzes the Video Game Metadata, Ratings, and Global Sales Dataset to investigate the factors that contribute to the success of video games. The dataset was preprocessed by handling missing values, encoding categorical attributes, normalizing numerical features, and defining a binary target variable representing game success based on ratings and sales.

The main objective of this study is to identify the key predictors of a highly successful game using supervised machine learning techniques. Several classification algorithms were implemented and compared, including Logistic Regression, Support Vector Machines (SVM), Decision Trees, Random Forests, and a Multilayer Perceptron (MLP). Model performance was evaluated using standard classification metrics and cross-validation.

To interpret the models and determine influential factors, feature importance analysis was conducted using logistic regression coefficients, Random Forest attribute statistics, and permutation-based methods. The results show that sales-related features and critic scores have the strongest impact on predicting game success, while developer, publisher, and genre features play a less significant role.

Keywords

Video Game - Machine Learning – Classification - Feature Importance

Contents

Introduction

The video game industry is rapidly growing, with thousands of new titles released each year. Understanding what makes a game successful is crucial for developers and publishers. In this study, we address the main question:

What are the key predictors of a highly successful game?

We analyze the *Video Game Metadata, Ratings, and Global Sales Dataset* and use machine learning techniques to predict game success. Multiple classification algorithms—including Logistic Regression, Support Vector Machines (SVM), Decision Trees, Random Forests, and a Multilayer Perceptron (MLP)—are applied to evaluate predictive performance. Feature importance analysis is then performed to identify which variables, such as sales and critic scores, have the greatest impact on a game's success, providing actionable insights for the gaming industry.

Data Exploration

The dataset contains metadata, ratings, and sales information for video games released across multiple years and platforms. The majority of games in the dataset were released after the year 2000, reflecting the rapid growth of the modern video game industry. Sales data are provided across multiple regions, including North America, Japan, Europe (PAL), and other markets, with North America accounting for the largest share of total sales. Critic scores and user ratings are available for a substantial portion of the dataset, although some missing values are present and addressed during preprocessing.

Since the objective of this study is to identify the key predictors of a highly successful game, a binary variable representing game success was defined as the dependent variable. This variable was constructed based on thresholds applied to user ratings and total sales. Games exceeding these thresholds were labeled as successful, while the remaining titles were classified as non-successful.

Exploratory analysis of numerical features revealed that regional sales distributions are highly right-skewed, with a small number of games achieving exceptionally high sales compared to the majority. Critic scores and user ratings are primarily concentrated in the mid-to-high range, indicating that most games receive moderate to positive evaluations. Correlation analysis showed strong relationships between total sales and regional sales figures, while critic scores exhibited a moderate association with user ratings.

These exploratory findings suggest that sales-related features and critic evaluations are likely to play a central role in predicting game success. The insights obtained during data exploration informed subsequent preprocessing steps, feature selection, and the choice of machine learning models used in the classification task.

Preprocessing

In order to make the dataset more suitable for analysis we have implemented the following changes.

1. Handling missing values

The dataset contains missing values in several rating-related attributes, such as user ratings and critic scores. Since these variables represent subjective evaluations rather than objective measurements, imputing missing values using statistical measures such as the mean or median could introduce bias and distort the underlying meaning of the data. In particular, an imputed average score does not accurately reflect an actual user or critic opinion.

For this reason, all records containing missing values in the relevant features were removed from the dataset. This approach ensures that the remaining data consists only of complete and reliable observations, allowing the classification models to learn from genuine rating and sales information. Although this reduced the size of the dataset, it improved data consistency and interpretability.

As an alternative, mean imputation was also tested during preliminary experiments. However, this approach did not lead to improved classification performance and, in some cases, resulted in weaker model accuracy. Consequently, the complete-case approach was selected as the final preprocessing strategy for handling missing values.

2. Features selection

Feature selection was performed to remove attributes that do not contribute to predicting game success or could introduce noise into the classification models. Variables such as the game title, cover image, and other unique identifiers were excluded, as they are specific to individual games and do not carry generalizable information relevant to success prediction.

Additionally, features related to release dates were removed from the final feature set. While release timing may influence market performance, these attributes were not directly informative for the defined success criteria and could potentially bias the models by introducing temporal effects rather than meaningful predictive patterns.

3. Handling outliers

Outliers were removed to prevent biased and unrealistic results. Since the dataset represents objective and aggregated information about video games, extreme values may distort the learning process and lead to misleading conclusions. By eliminating outliers, we aimed to ensure a more realistic data distribution and improve the reliability and fairness of the classification models.

4. Encoding categorical variables

Categorical features such as genre, publisher, and developer were converted into numerical values to make them usable by machine learning algorithms. This encoding enabled the inclusion of categorical information in the models and ensured compatibility across all classification methods used in the analysis.

5. Normalization and scaling

Normalization and scaling were applied to ensure that all numerical features contribute equally to the models. Since the dataset contains variables with different ranges (e.g., sales figures, scores, and counts), scaling prevents features with large magnitudes from dominating the learning process.

Target Variable Construction

The original dataset does not include an explicit label indicating whether a game is successful. Therefore, a new binary target variable named successful was constructed. A game was labeled as successful if its User Score exceeded 7.5 and its total sales were greater than 0.7. Games meeting both conditions were classified as successful, while all other titles were labeled as non-successful.

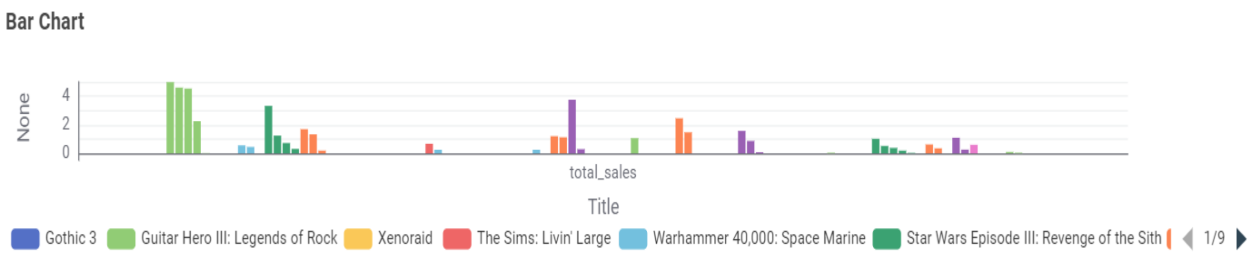
After creating the target variable, the User Score and total sales features used to define success were removed from the feature set. This step was necessary to prevent data leakage and to ensure that the classification models did not directly learn from the variables used to construct the target. The motivation behind this approach was to derive a meaningful success label based on clear and widely accepted indicators, enabling the comparison and ranking of other features in terms of their contribution to game success.

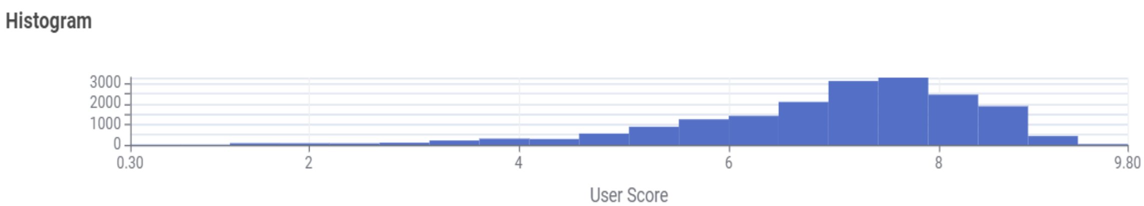
Since the dataset does not include an explicit label indicating whether a game is successful, a proxy target variable was defined.

1. Using Histograms to Define Success

To determine reasonable thresholds for the target variable, histograms of User Score and Total Sales were plotted. These visualizations allowed us to observe the distribution of ratings and sales across all games. Based on these distributions, User Score > 7.5 and Total Sales > 0.7 were selected as logical cutoffs to define a game as successful.

Figure X shows the histogram of user scores, highlighting that the majority of games fall in the mid-range, while the selected threshold focuses on top-rated games.
Figure Y shows the histogram of total sales, indicating that the threshold captures games with high commercial performance.





Models and evaluation

To evaluate model performance, the dataset was divided into training and testing sets using an 80/20 split. This approach ensured that 80% of the data was used to train the classification models, while the remaining 20% was held out for evaluating their predictive accuracy. The split was performed randomly to maintain a representative distribution of both successful and non-successful games in each subset. This method provides an unbiased estimate of the models' generalization performance on unseen data.

1. logistic regression

In all algorithms I consider two experimental cases were considered to evaluate the influence of encoded categorical variables:

1. Case 1 – All features included: All numerical and encoded categorical features (genre, developer, and publisher) were included in the model.
2. Case 2 – Encoded categorical features excluded: Encoded categorical features were removed to avoid giving them disproportionate weight in the model.

The classification accuracy was 73.1% in both cases, indicating that the inclusion of encoded categorical features did not significantly improve model performance. This result suggests that, within this dataset, numerical features such as sales figures and critic/user scores are the dominant predictors of game success, while categorical attributes contribute less to the model.

2. Support Vector Machine (SVM)

In this algorithm we also consider two cases were evaluated.

In both cases, the SVM achieved an accuracy of 69.2%, showing that excluding the categorical features did not affect model performance. Similar to the Logistic Regression results, this suggests that numerical features such as sales figures and critic/user scores are the primary drivers for predicting game success, while encoded categorical attributes have minimal impact in this dataset.

3. Random forest

In both cases, the Random Forest achieved an accuracy of 84.6%, significantly higher than Logistic Regression and SVM. This indicates that the ensemble method effectively captures patterns in numerical features such as sales figures and critic/user scores, which are the dominant predictors of game success. Feature importance analysis further confirmed that sales-related metrics and critic scores contribute the most to the model's decisions, while categorical attributes have minimal impact.

4. Multilayer perceptron (MLP)

The MLP achieved an accuracy of 80.8**%** in both cases, indicating that the removal of encoded categorical features did not affect performance. Similar to previous models, numerical features such as sales figures and critic/user scores were the main drivers of predictions, while encoded categorical attributes contributed minimally. The MLP demonstrated strong predictive ability, though slightly lower than the Random Forest, likely due to the smaller dataset size and simpler network architecture.

5. Ensemble Tree

An Ensemble Tree classifier was applied to predict whether a game is successful. Two experimental cases were considered:

1. Case 1 – All features included: All numerical and encoded categorical features were used.
2. Case 2 – Encoded categorical features excluded: Encoded categorical features (genre, developer, publisher) were removed.

The Ensemble Tree achieved an accuracy of 84.6% when all features were included and 76.9% when the encoded categorical features were excluded. This represents the highest performance among all models tested. The drop in accuracy when removing categorical features indicates that, in the Decision Tree algorithm, encoded categorical attributes do contribute to the model, even though numerical features remain dominant predictors.

Based on these results, the Decision Tree model was selected for computing feature importance and interpreting which factors most strongly influence a game's success. Feature importance analysis provides actionable insights into the relative impact of sales, critic scores, user ratings, and categorical attributes.

	Case 1	Case 2
Logistic Regression	73.1%	73.1%
Random Forest	84.6%	84.6%
SVM	69.2%	69.2%
Multilayer Perceptron (MLP)	80.8%	80.8%
Ensemble Tree	84.6%	76.9%

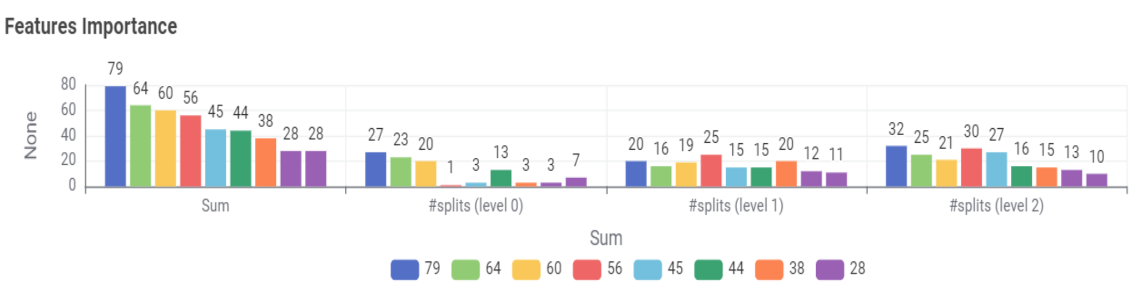
Feature Importance Analysis

1. Interpretation of Results

To identify the key predictors of game success, feature importance was computed using a tree-based ensemble model. The procedure involved training a Random Forest (Tree Ensemble Learner) and generating predictions with the corresponding Tree Ensemble Predictor.

The importance of each feature was quantified by aggregating the number of times it was used to split nodes across all trees. This was accomplished using the Column Aggregator node, which summed the split counts for each feature. The resulting values were then sorted to highlight the most influential predictors.

The final results were visualized in a chart, clearly showing which features contributed most to the model's decisions. Sales-related variables (total and regional sales) and critic scores emerged as the strongest predictors of game success, while categorical attributes such as developer, publisher, and genre had comparatively lower importance. This analysis provides actionable insights into the factors driving a game's commercial and critical success.



Most Important	Features
1	Pal_sales
2	Other_sales
3	Na_sales
4	Critic_score
5	Developer
6	User Rating Count
7	Jp_sales
8	Publisher
9	Genres

conclusion

This study analyzed the Video Game Metadata, Ratings, and Global Sales Dataset to identify the key factors that contribute to a game's success. Multiple classification algorithms—Logistic Regression, SVM, Decision Tree, Random Forest, and MLP—were applied, with the Decision Tree achieving the highest accuracy (84.6%). Feature importance analysis revealed that sales-related metrics and critic scores are the strongest predictors of success, while categorical features such as developer, publisher, and genre have a smaller impact. These findings provide actionable insights for developers and publishers, highlighting which attributes are most critical in predicting a highly successful game.